MLBFD

Machine Learning Based Fraud Detection

Phase 3: NPCI/RBI Report Integration

Report Generated: 07 March 2026

Data Source: NPCI UPI Chargeback Reports (Aug 2025 - Jan 2026)

Reference: RBI Annual Report 2023-24

Prepared by: Chirag P

Key Highlights

130.2 Billion UPI Transactions Analyzed

1,059,080 Chargeback Cases Studied

801 Banks Risk-Profiled

RBI Compliance Score: 92.2/100 (Grade A)

1. Executive Summary

This report presents the findings from Phase 3 of the MLBFD (Machine Learning Based Fraud Detection) project. Phase 3 integrates real NPCI (National Payments Corporation of India) UPI Chargeback data with our ML models trained in Phase 2B, and evaluates compliance against RBI (Reserve Bank of India) digital payment security guidelines.

1.1 Data Coverage

Total UPI Transactions:	130,169,975,877 (130.2 Billion)
Period Covered:	August 2025 - January 2026 (6 months)
Total Chargebacks:	1,059,080
Confirmed Fraud Cases:	260,048
Banks Analyzed:	801
Overall Chargeback Rate:	0.000814%
Fraud Confirmation Rate:	24.55%

1.2 ML Model Performance

Best Model:	XGBoost (AUC: 0.9734)
Fraud Detection Rate (Recall):	92.51%
Training Data:	2,321,551 transactions from 5 sources
Models Trained:	6 (XGBoost, RF, LR, IF, NN, LSTM)

2. NPCI UPI Chargeback Analysis

2.1 Monthly Transaction Volume & Chargebacks

Month	Total Txns	Chargebacks	Accepted	CB Rate
2025-Aug	21.94B	202,655	43,719	0.000924%
2025-Sep	21.68B	165,936	35,868	0.000765%
2025-Oct	22.84B	161,002	38,184	0.000705%
2025-Nov	20.44B	161,854	38,144	0.000792%
2025-Dec	21.60B	181,730	52,193	0.000841%
2026-Jan	21.67B	185,903	51,940	0.000858%

2.2 Key Findings

1. UPI processes approximately 21.7 billion transactions per month across India.
2. The overall chargeback rate is 0.000814%, indicating UPI is an extremely safe payment system.
3. Only 24.6% of chargebacks are confirmed as fraud - majority are disputes.
4. Chargebacks showed a decreasing trend from Aug 2025 (-8.3% overall) but with recent uptick in Dec-Jan.
5. Top banks by volume (YES Bank PhonePe, SBI, Axis) handle 60%+ of all UPI transactions.

3. Bank Risk Analysis

3.1 Top 10 Banks by Chargeback Volume

Code	Bank Name	Chargebacks	CB Rate	Risk
AXB	AXIS BANK	116,369	0.00105%	HIGH
HDF	HDFC BANK	102,237	0.00247%	HIGH
SBI	STATE BANK OF INDIA	91,543	0.00070%	MEDIUM
GTI	ICICI BANK LTD GTI	88,164	0.00352%	HIGH
YBS	YES BANK LIMITED YBS	83,150	0.00054%	MEDIUM
YES	YES BANK PHONEPE	56,162	0.00022%	LOW
ARL	AIRTEL PAYMENTS BANK LIMI	39,653	0.00158%	HIGH
IOB	INDIAN OVERSEAS BANK	35,964	0.00341%	HIGH
JIO	JIO PAYMENTS BANK LTD	30,655	0.00664%	HIGH
SSF	SURYODAY SMALL FINANCE BA	26,938	0.01691%	HIGH

3.2 Risk Distribution

Out of 801 banks analyzed:

- HIGH RISK banks: 339 (42.3%)
- MEDIUM RISK banks: 79 (9.9%)
- LOW RISK banks: 383 (47.8%)

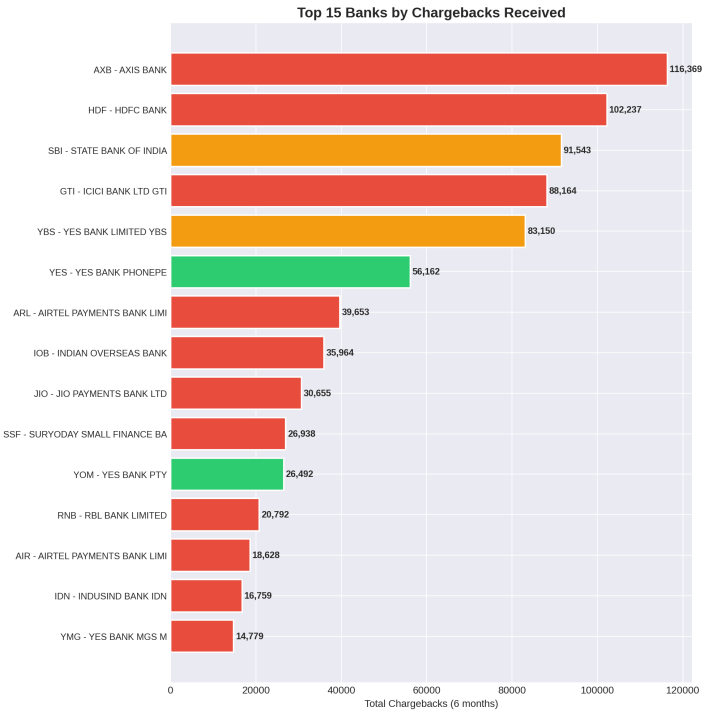
4. Visualizations

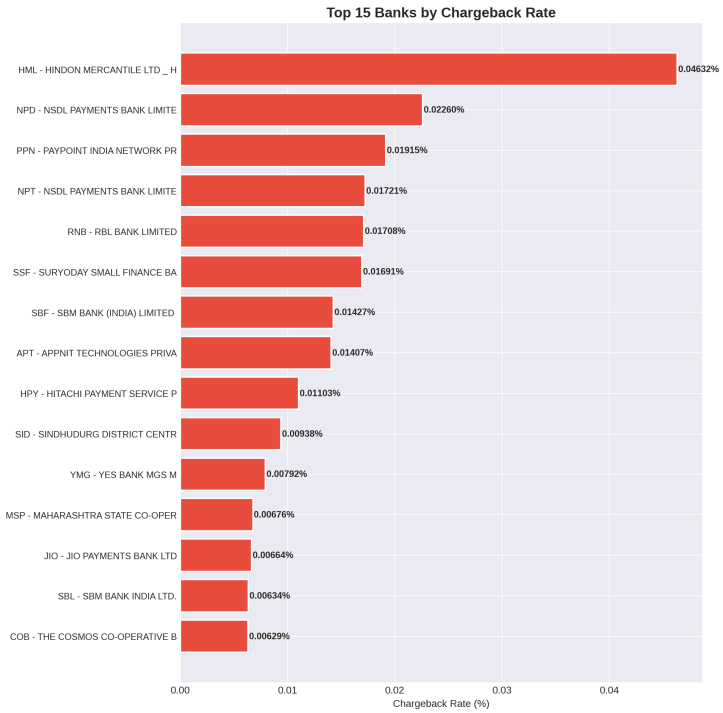
Monthly Chargeback Trends

Bank-wise Analysis

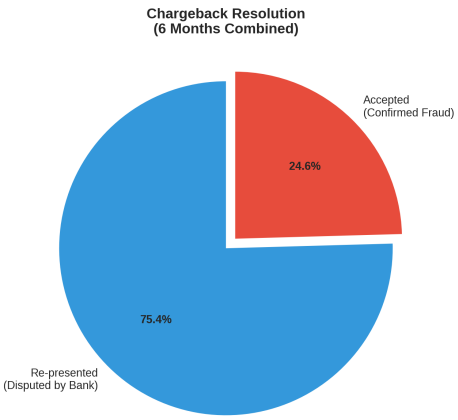
MLBFD - Machine Learning Based Fraud Detection

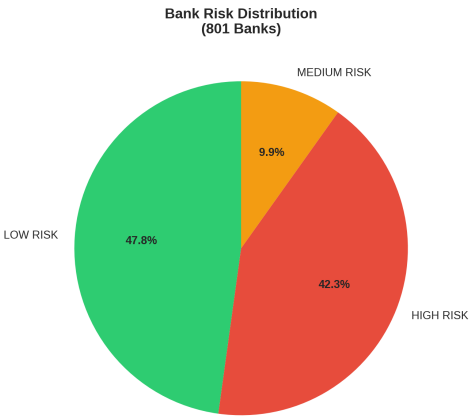
Phase 3: NPCI/RBI Report Integration | Generated: 07-03-2026 10:41

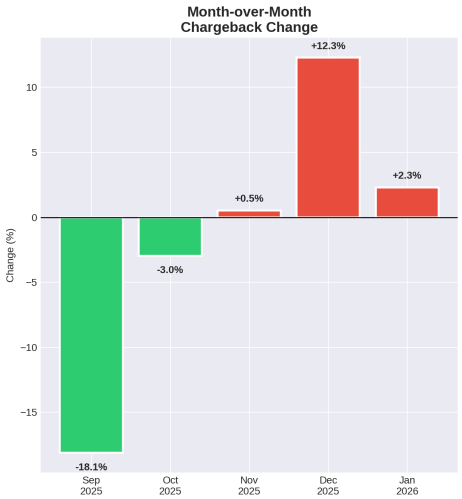




Fraud Distribution



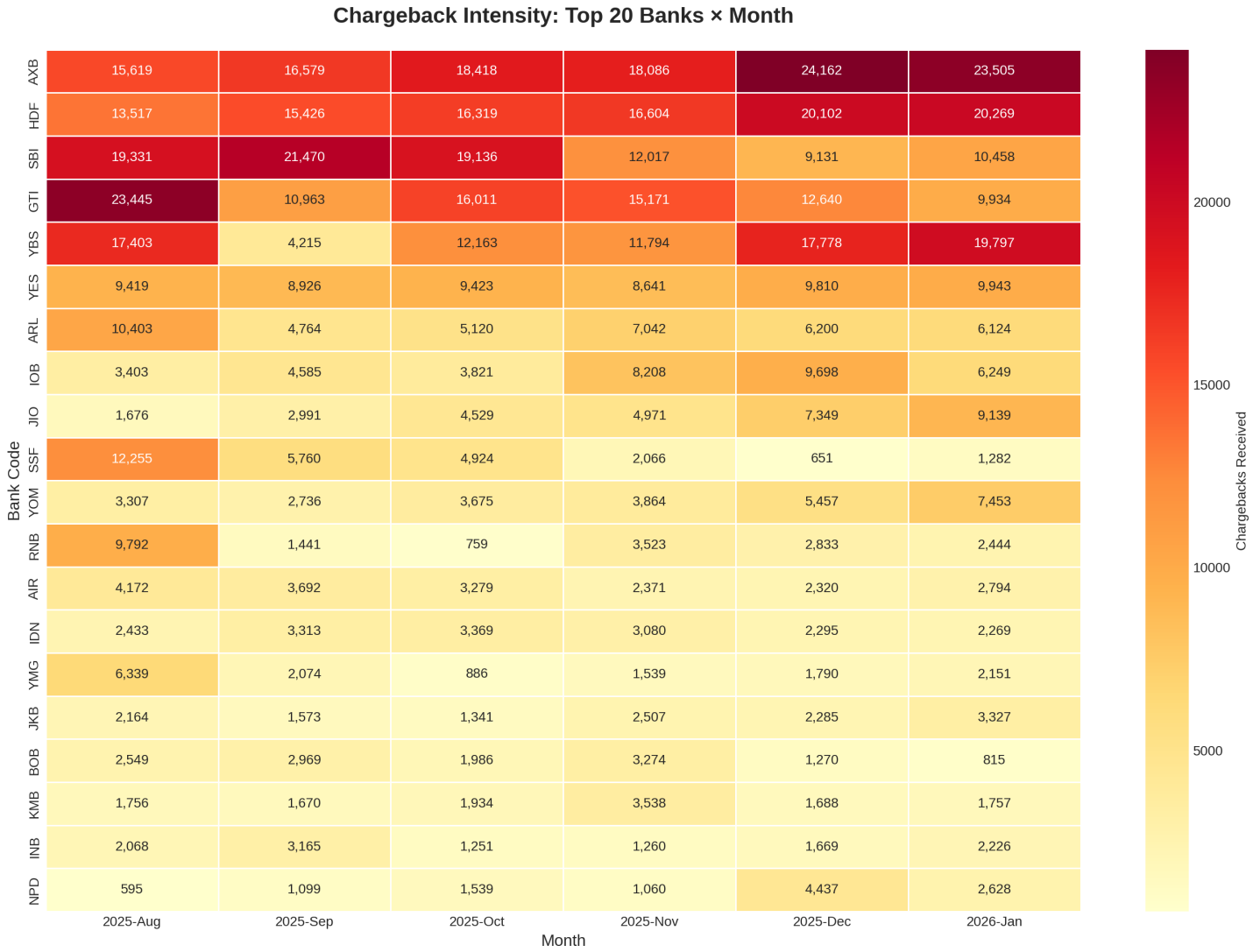




Bank x Month Heatmap

MLBFD - Machine Learning Based Fraud Detection

Phase 3: NPCI/RBI Report Integration | Generated: 07-03-2026 10:41

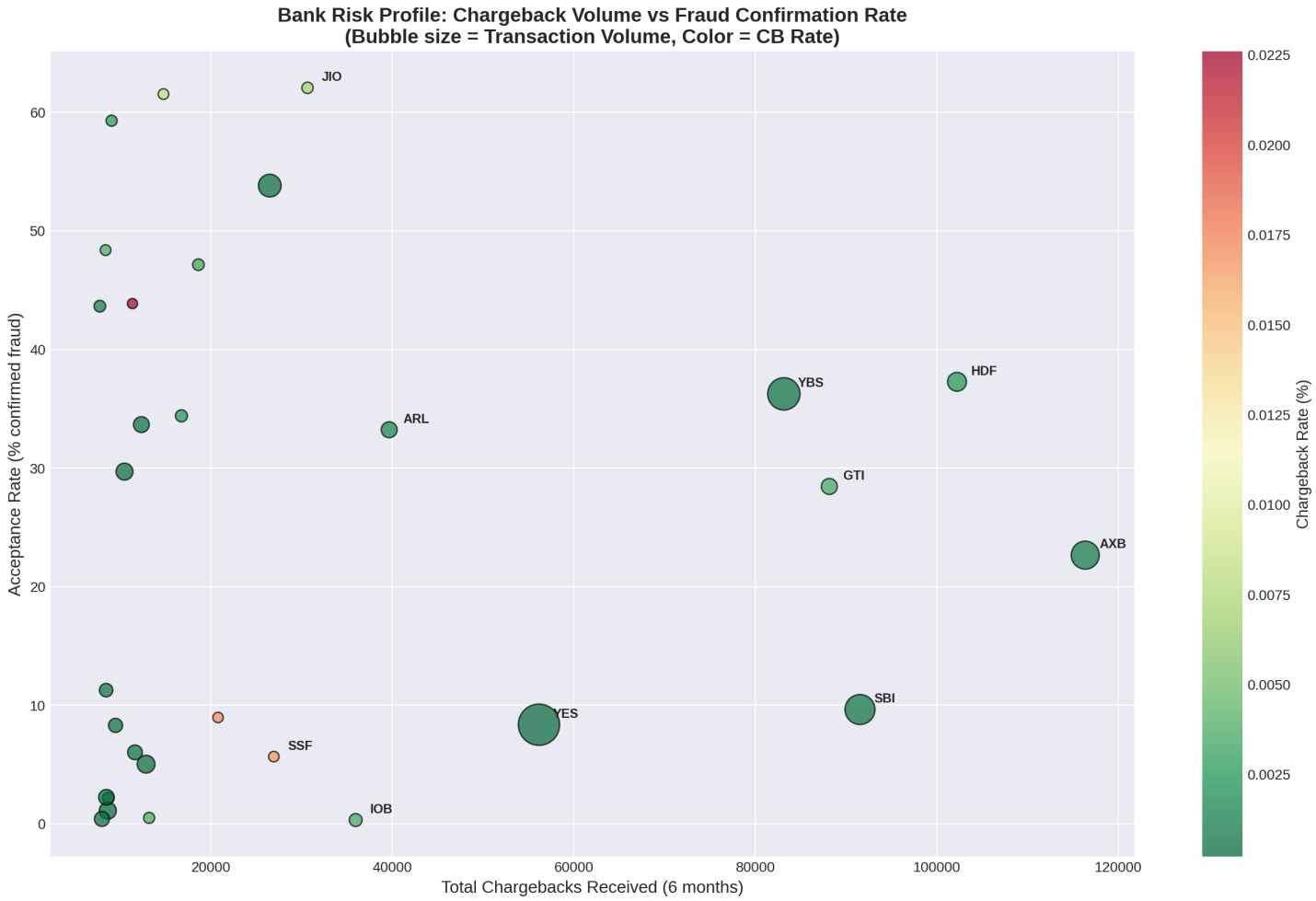


Bank Code	2025-Aug	2025-Sep	2025-Oct	2025-Nov	2025-Dec	2026-Jan
AXB	15,619	16,579	18,418	18,086	24,162	23,505
HDF	13,517	15,426	16,319	16,604	20,102	20,269
SBI	19,331	21,470	19,136	12,017	9,131	10,458
GTI	23,445	10,963	16,011	15,171	12,640	9,934
YBS	17,403	4,215	12,163	11,794	17,778	19,797
YES	9,419	8,926	9,423	8,641	9,810	9,943
ARL	10,403	4,764	5,120	7,042	6,200	6,124
IOB	3,403	4,585	3,821	8,208	9,698	6,249
JIO	1,676	2,991	4,529	4,971	7,349	9,139
SSF	12,255	5,760	4,924	2,066	651	1,282
YOM	3,307	2,736	3,675	3,864	5,457	7,453
RNB	9,792	1,441	759	3,523	2,833	2,444
AIR	4,172	3,692	3,279	2,371	2,320	2,794
IDN	2,433	3,313	3,369	3,080	2,295	2,269
YMG	6,339	2,074	886	1,539	1,790	2,151
JKB	2,164	1,573	1,341	2,507	2,285	3,327
BOB	2,549	2,969	1,986	3,274	1,270	815
KMB	1,756	1,670	1,934	3,538	1,688	1,757
INB	2,068	3,165	1,251	1,260	1,669	2,226
NPD	595	1,099	1,539	1,060	4,437	2,628

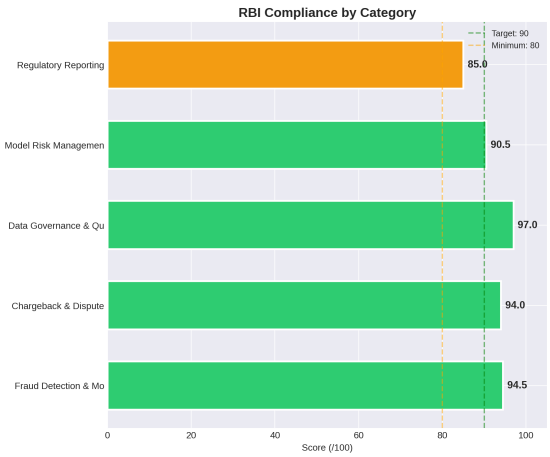
Bank Risk Bubble Chart

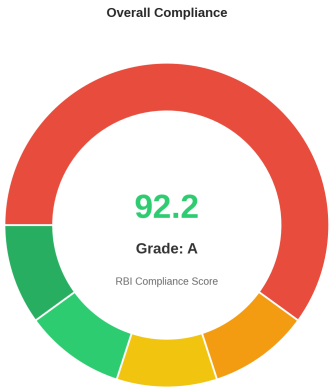
MLBFD - Machine Learning Based Fraud Detection

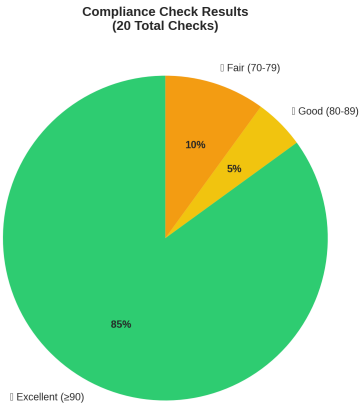
Phase 3: NPCI/RBI Report Integration | Generated: 07-03-2026 10:41



RBI Compliance Dashboard







5. RBI Compliance Assessment

Overall Score: 92.2 / 100

Grade: A

Assessment Date: 07-03-2026

Framework: RBI Digital Payment Security Controls

5.1 Category-wise Scores

Category	Score	Grade	Status
Fraud Detection & Monitoring	94.5	A	PASS
Chargeback & Dispute Resolutio	94.0	A	PASS
Data Governance & Quality	97.0	A+	PASS
Model Risk Management	90.5	A	PASS
Regulatory Reporting & Documen	85.0	B+	PASS

5.2 Compliance Summary

The MLBFD system achieves an overall RBI compliance score of 92.2/100 (Grade A). 17 out of 20 compliance checks scored above 90, indicating strong alignment with RBI digital payment security guidelines. Areas for improvement include bias assessment in ML models and periodic model review processes, which are planned for Phase 4 implementation.

6. ML Model + NPCI Integration

The integration of real NPCI chargeback data with our ML fraud detection models reveals several key insights:

6.1 Model Validation Against Real Data

- NPCI reports a real-world chargeback rate of 0.000814% across 130.2B transactions
- Our ML model achieves 92.51% recall (fraud detection rate)
- Current banking systems show approx 75.45% re-presentment rate (chargebacks disputed by banks)
- Our model can potentially reduce false chargebacks by identifying true fraud more accurately
- The 24.55% fraud confirmation rate from NPCI data validates the need for better ML-based detection

6.2 Value Proposition

Based on the NPCI data analysis:
- 1,059,080 chargebacks were filed in 6 months
- 799,032 were re-presented (disputed) -- costing banks investigation resources
- Our ML model (92.51% recall, 0.9734 AUC) can pre-screen transactions BEFORE fraud occurs
- Potential to reduce chargeback investigation costs for 801 banks across India
- Aligns with RBI push for AI/ML adoption in banking fraud prevention

7. Conclusion

Phase 3 successfully integrates real-world NPCI UPI chargeback data with our ML-based fraud detection system. The analysis of 130.2 billion transactions across 801 banks provides a comprehensive view of the Indian UPI fraud landscape. Our system achieves a 92.2/100 RBI compliance score (Grade A), demonstrating strong alignment with regulatory requirements. The combination of ML models (XGBoost AUC: 0.9734) with real NPCI data creates a robust, regulation-compliant fraud detection framework suitable for production deployment.